

Feynman Rules : $\langle f | S | i \rangle \sim \langle \Omega | T \{ \phi_1 \dots \phi_n \} | \Omega \rangle$

$$[\phi, \phi'] = 0, \quad [\phi(\vec{x}, t), \partial_t \phi(\vec{x}', t)] = i\hbar \delta^3(\vec{x} - \vec{x}')$$

• Assume ϕ satisfy E.L. equations, we now how to calculate $\langle \Omega | T \{ \phi_1 \phi_2 \} | \Omega \rangle$ in a free theory

$$(\Box + m^2) \langle \Omega | T \{ \phi(x) \phi(x') \} | \Omega \rangle = \langle \Omega | T \{ (\Box + m^2) \phi(x) \phi(x') \} | \Omega \rangle - i\hbar \delta^4(x - x')$$

Proof: $\phi(x) \equiv \phi, \quad \phi(x') \equiv \phi'$

$$\begin{aligned} \partial_t \langle \Omega | T \{ \phi \phi' \} | \Omega \rangle &= \partial_t \left[\langle \Omega | \phi \phi' | \Omega \rangle \theta(t - t') + \langle \Omega | \phi' \phi | \Omega \rangle \theta(t' - t) \right] \\ &= \langle \Omega | \partial_t \phi \phi' | \Omega \rangle \theta(t - t') + \langle \Omega | \phi \phi' | \Omega \rangle \partial_t \theta(t - t') \\ &\quad + \langle \Omega | \phi' \partial_t \phi | \Omega \rangle \theta(t' - t) - \langle \Omega | \phi' \phi | \Omega \rangle \partial_t \theta(t' - t) \end{aligned}$$

$$= \langle \Omega | T \{ \partial_t \phi \phi' \} | \Omega \rangle + \delta(t - t') \langle \Omega | \phi \phi' - \phi' \phi | \Omega \rangle$$

∴

$$\partial_t \langle \Omega | T \{ \phi \phi' \} | \Omega \rangle = \langle \Omega | T \{ \partial_t \phi \phi' \} | \Omega \rangle + \delta(t - t') \langle \Omega | [\phi, \phi'] | \Omega \rangle$$

$$\delta(t - t') \Rightarrow t = t'$$

and

$$[\phi, \phi']_t = 0$$

$$\begin{aligned} \partial_t^2 \langle \Omega | T \{ \phi \phi' \} | \Omega \rangle &= \langle \Omega | T \{ \partial_t^2 \phi \phi' \} | \Omega \rangle \\ &\quad + \delta(t - t') \langle \Omega | [\partial_t \phi, \phi'] | \Omega \rangle \\ &= \langle \Omega | T \{ \partial_t^2 \phi \phi' \} | \Omega \rangle + \delta(t - t') (-i\hbar \delta^3(\vec{x} - \vec{x}')) \end{aligned}$$

* For a free scalar field theory w/

$$(\Box + m^2)\phi_0 = 0$$

$$(\Box + m^2) D_F(x, y) = -i \delta^4(x - y)$$

* In general, for FSFT:

$$\Box_x \langle \phi_x \phi_1 \dots \phi_n \rangle = \langle \Box_x \phi \phi_1 \dots \phi_n \rangle - i \delta^4(x - x_j) \langle \phi_1 \dots \phi_{j-1} \phi_{j+1} \dots \phi_n \rangle$$

* For scalar interactions

$$\mathcal{L} = -\frac{1}{2} \phi (\Box + m^2) \phi + \mathcal{L}_{int}[\phi]$$

then

$$(\Box + m^2) \phi - \mathcal{L}'_{int}[\phi] = 0$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0$$

$$-\frac{1}{2} \phi \partial^\sigma \partial_\sigma \phi - \frac{1}{2} m^2 \phi^2$$

$$\partial_\mu \left(-\frac{1}{2} \phi \partial^\sigma \partial_\sigma \delta(x - y) \right) = \frac{1}{2} \Box \phi + m^2 \phi$$

$$\partial_\mu \left(\int \frac{1}{2} \phi \partial^\mu \delta(x - y) \right)$$

I.B.P

$$\therefore + \Box \phi + m^2 \phi - \mathcal{L}'_{int}[\phi] = 0$$

✓

$$(\Box_x + m^2) \langle \phi_x \phi_1 \dots \phi_n \rangle = \langle \mathcal{L}'_{int}[\phi] \phi_1 \dots \phi_n \rangle - i \delta^4(x - x_j) \langle \phi_1 \dots \phi_{j-1} \phi_{j+1} \dots \phi_n \rangle$$

(Schwinger-Dyson)

Equations

Position-Space Feynman Rules:

• Free theory with $\mathcal{L}_{int} = 0$

$$(\Box + m^2) \langle \phi, \phi_2 \rangle = -i \delta^4(x_1 - x_2)$$

$$\langle \phi, \phi_2 \rangle = D_{12}$$

$$\left. \begin{aligned} \delta_{x_i} &\equiv \delta(x - x_i) \\ D_{ij} &= D(x_i - x_j) \end{aligned} \right\} m = 0 \text{ (massless)}$$

$$\boxed{\Box_x D_{x_1} = -i \delta_{x_1}}$$

$$\begin{aligned} \langle \phi, \phi_2 \rangle &= \int d^4x \delta_{x_1}^x \langle \phi_x \phi_2 \rangle = i \int d^4x \Box_x D_{x_1} \langle \phi_x \phi_2 \rangle \\ &= i \int d^4x D_{x_1} \Box_x \langle \phi_x \phi_2 \rangle \end{aligned}$$

$\therefore$

$$\langle \phi, \phi_2 \rangle = i \int d^4x D_{x_1} \Box_x \langle \phi_x \phi_2 \rangle$$

S.D. Eq's:

$$\langle \phi, \phi_2 \rangle = i \int d^4x D_{x_1} (-i \delta_{x_2}) = D_{21} = D_{12}$$

$$\begin{aligned} \bullet \langle \phi_1 \dots \phi_4 \rangle &= \int d^4x \delta_{x_1} \langle \phi_x \phi_2 \phi_3 \phi_4 \rangle \\ &= i \int d^4x \Box_x D_{x_1} \langle \phi_x \phi_2 \phi_3 \phi_4 \rangle \\ &= i \int d^4x D_{x_1} \Box_x \langle \phi_x \phi_2 \dots \phi_4 \rangle \end{aligned}$$

$$\begin{aligned} &= i \int d^4x D_{x_1} (-i) (\delta_{x_2} \langle \phi_3 \phi_4 \rangle + \delta_{x_3} \langle \phi_2 \phi_4 \rangle + \delta_{x_4} \langle \phi_2 \phi_3 \rangle) \\ &= D_{12} \langle \phi_3 \phi_4 \rangle + D_{13} \langle \phi_2 \phi_4 \rangle + D_{14} \langle \phi_2 \phi_3 \rangle \\ &= D_{12} D_{34} + D_{13} D_{24} + D_{14} D_{23} \end{aligned}$$



Let us now consider an interaction in the Lag.

$$\mathcal{L} = \frac{1}{2} \phi^2 - \frac{g}{3!} \phi^3$$

$$\langle \phi, \phi_2 \rangle = \int d^4x \delta_{1x} \langle \phi_x \phi_2 \rangle$$

$$\square \langle \rangle = \langle \mathcal{L}'_{int} \rangle - i t_h \delta$$

$$\delta = i(\langle \mathcal{L}'_{int} \rangle - \square \langle \rangle)$$

$$\begin{aligned} \langle \phi, \phi_2 \rangle &= i \int d^4x \square_x D_{x1} \langle \phi_x \phi_2 \rangle \\ &= i \int d^4x D_{x1} \square_x \langle \phi_x \phi_2 \rangle \end{aligned}$$

$$= i \int d^4x D_{x1} \left(\langle \frac{g}{2} \phi_x^2 \phi_2 \rangle - i \delta_{x2} \right)$$

• Introduce a new integration over $\delta_{2y} \equiv i \square_y D_{y2}$

$$\bullet i \int d^4x \int d^4y \delta_{2y} D_{x1} \langle \frac{g}{2} \phi_x^2 \phi_y \rangle$$

$$\begin{aligned} &= - \int d^4x d^4y D_{x1} \square_y D_{y2} \langle \phi_x^2 \phi_y \rangle \\ &= - \int d^4x d^4y D_{x1} D_{y2} \square_y \langle \phi_x^2 \phi_y \rangle \end{aligned}$$

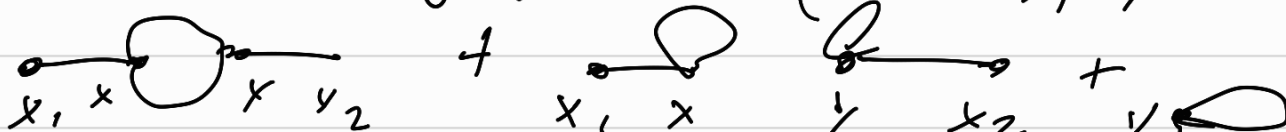
$$\square_y \langle \phi_x^2 \phi_y \rangle = \frac{g^2}{4} \langle \phi_x^2 \phi_y^2 \rangle - i g \langle \phi_x \rangle$$

$$\langle \phi_x^2 \phi_y^2 \rangle = 2 D_{xy}^2 + D_{xx} D_{yy} + \mathcal{O}(g)$$

$$\begin{aligned} \langle \phi_x \rangle &= \int d^4y \delta_{1x} \langle \phi_y \rangle \\ &= i \int d^4y \square_y D_{yx} \langle \phi_y \rangle \end{aligned}$$

$$\begin{aligned} &= i \int d^4y D_{xy} \square_y \langle \phi_y \rangle \\ &= i \int d^4y D_{xy} \frac{g}{2} \langle \phi_y^2 \rangle \\ &= i \frac{g}{2} \int d^4y D_{xy} D_{yy} + \mathcal{O}(g^2) \end{aligned}$$

$$\langle \phi, \phi_2 \rangle = D_{12} - g^2 \int d^4x d^4y \left(\frac{1}{2} D_{1x} D_{xy}^2 D_{y2} + \frac{1}{4} \dots \right)$$





Hamiltonian Derivation:

• time ind. perf. theory

• Instead of assuming ϕ satisfies the E-L. eq's we instead assume its dynamics are determined by H

$$i\partial_t \phi = [\phi, H] = \phi H - H\phi$$

$$(H + i\partial_t)\phi = \phi H$$

* By being in the Heisenberg pic:

$$\phi(\vec{x}, t) = S(t, t_0)^\dagger \phi(\vec{x}) S(t, t_0)$$

$S(t, t_0)$: S-Matrix/time evolution op.

$$i\partial_t (S^\dagger \phi S) = S^\dagger \phi S H - H S^\dagger \phi S$$

$$i\partial_t S^\dagger \phi S + iS^\dagger \phi \partial_t S = S^\dagger \phi S H - H S^\dagger \phi S$$

$$iS\partial_t S^\dagger \phi S + i\phi(\partial_t S) = \phi S H - S H S^\dagger \phi S$$

$$iS\partial_t S^\dagger \phi + i\phi(\partial_t S)S^\dagger = \phi S H S^\dagger - S H S^\dagger \phi$$

$$S^\dagger S = \mathbb{1} \Rightarrow \partial_t S S^\dagger = -S \partial_t S^\dagger$$

$$\Rightarrow iS\partial_t S^\dagger \phi - i\phi S\partial_t S^\dagger = \dots$$

$$= [\phi, iS\partial_t S^\dagger] = [\phi, S H S^\dagger]$$

$$\therefore -iS\partial_t S^\dagger = S H S^\dagger$$

... in text its

$$i\partial_t S(t, t_0) = H(t) S(t, t_0)$$

$$H(t) = H_0 + V(t)$$

Note this is the full Hamiltonian and $\mathcal{H}$ is the density $\mathcal{H} = \frac{g}{3} \phi^3$ is the density s.t. $H_V = \int d^3x \frac{g}{3} \phi(x, t)^3$

* Interaction Picture: ϕ evolves with H_0 only
 $\phi_{0I}(x, t) = e^{iH_0(t-t_0)} \phi(x) e^{-iH_0(t-t_0)} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\epsilon_p}} (a_p e^{ipx} + \dots)$

$$\phi(x) = e^{-iH_0(t-t_0)} \phi_0(\vec{x}, t)_I e^{iH_0(t-t_0)}$$

$$\phi(\vec{x}, t) = S^\dagger(t, t_0) e^{-iH_0(t-t_0)} \phi_0(\vec{x}, t) e^{iH_0(t-t_0)} S(t, t_0) = U^\dagger \phi_0 U$$

$$U(t, t_0) \equiv e^{iH_0(t-t_0)} S(t, t_0)$$

U : Relates Heisenberg ops to free fields at the same time.

$$\begin{aligned} * i\partial_t U &= ie^{iH_0(t-t_0)} (iH_0) S(t, t_0) + e^{iH_0(t-t_0)} H(t) S(t, t_0) \\ &= e^{iH_0(t-t_0)} (H(t) - H_0) S(t, t_0) \\ &= e^{iH_0(t-t_0)} V(t) S(t, t_0) \end{aligned}$$

* Express $V(t)$ in the interaction picture

$$U_I(t) \equiv e^{-iH_0 t} V e^{iH_0 t} \Rightarrow V = e^{iH_0 t} U_I e^{-iH_0 t}$$

$$i\partial_t U - V_I e^{iH_0(t-t_0)} S(t, t_0)$$

$$\therefore i\partial_t U(t, t_0) = V_I(t) U(t, t_0)$$

Formally the solution is

$$U(t, t_0) = T \left\{ \exp \left[-i \int_{t_0}^t dt' V_I(t') \right] \right\}$$

• Time ordering allows $V_I(t_i)$ and $V_I(t_j)$ to commute.

B.C.'s: $U(t, t) = I$

$$U_{21} \equiv U(t_2, t_1) = T \left\{ e^{-i \int_{t_1}^{t_2} dt' V_I(t')} \right\}$$

$$U_{21} U_{12} = I$$

$$U_{21}^{-1} = U_{21}^\dagger = U_{12}$$

since

$$U_{21}^\dagger = e^{i \int_{t_1}^{t_2} dt' V_I} = e^{-i \int_{t_2}^{t_1} dt' V_I(t')}$$

$$U_{32} U_{21} = U_{31}$$

$$U_{31} U_{12} = U_{32} \quad \text{etc} \dots$$

Since

$$\phi(\vec{r}, t) = U^\dagger(t, t_0) \phi_0(\vec{r}, t_0) U(t, t_0)$$

$$\phi(x_1) = U_{10}^\dagger \phi_0(x_1, t_1) U_{10}$$

Vacuum Matrix elements:

$|\Omega\rangle$ ann. by $a_{\mathbf{p}}|t\rangle$ at $t \rightarrow \infty$ as well as $S(t, t_0)|\Omega\rangle$ where we have evolved $|\Omega\rangle$ to the state $e^{iH_0(t-t_0)}|0\rangle$

t_0 : free and int. theory are equal

$$|\Omega\rangle = N_i \lim_{t \rightarrow -\infty} S^\dagger e^{iH_0(t-t_0)}|0\rangle$$

$$U = e^{iH_0(t-t_0)} S \Rightarrow U^\dagger = S^\dagger e^{-iH_0(t-t_0)}$$

$$\lim_{t \rightarrow \infty} S^\dagger(t, t_0) e^{-iH_0(t_0-t)}|0\rangle = U_{0-\infty}$$

$$\therefore |\Omega\rangle = N_i \lim_{t \rightarrow -\infty} U_{0-\infty}|0\rangle$$

$$\langle \Omega| = N_f \langle 0| U_{00} \quad t_1 > t_2 > \dots > t_n$$

$$\therefore \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle$$

$$= N_f N_i \langle 0 | U_{00} (U_{01} \phi_0(x_1) U_{10}) U_{02} \phi_0(x_2) U_{20} \dots \\ \dots U_{0n} \phi_0(x_n) U_{n0} U_{0-\infty} | 0 \rangle$$

$$= N_f N_i \langle 0 | U_{0\infty} \phi_0(x_1) U_{12} \phi_0(x_2) U_{23} \dots \\ \phi_0(x_n) U_{n-\infty} | 0 \rangle$$

$$= N_f N_i \langle 0 | T (\dots) | 0 \rangle$$

$$= N_f N_i \langle 0 | \phi_0(x_1) \dots \phi_0 U_{\infty, -\infty} | 0 \rangle$$

Normalize s.t. $\langle \Omega | \Omega \rangle = 1$

$$\therefore N_i N_f \langle 0 | U_{\infty, -\infty} | 0 \rangle = 1$$

$$N_i N_f = \exp[-i \int_{-\infty}^{\infty} dt' U_{\pm}(t')]$$

$\therefore$

$$-i \int_{-\infty}^{\infty} dt' U_{\pm}(t')$$

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \frac{\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} \exp(-i \int_{-\infty}^{\infty} dt' U_{\pm}(t')) | \Omega \rangle}{\langle \Omega | T \{ \exp(-i \int_{-\infty}^{\infty} dt' U_{\pm}(t')) \} | \Omega \rangle}$$

Momentum space Feynman Rules: $e^{i \int d^4x \mathcal{L}_{int}}$

$$\tau_1 \equiv \text{diagram} = \frac{-g^2}{2} \int d^4x d^4y D_{1x} D_{xy}^2 D_{y2}$$

symmetry factor

$$D_{ij} = \int \frac{d^4p}{(2\pi)^4} \left(\frac{i}{p^2 + i\epsilon} \right) e^{ip(x_i - x_j)}$$

$$\tau_1 = \frac{-g^2}{2} \int d^4x d^4y \left(\prod_{i=1}^4 \int \frac{d^4p_i}{(2\pi)^4} \left(\frac{i}{p_i^2 + i\epsilon} \right) \right) \frac{e^{ip_1(x_1 - x)} e^{ip_2(x - x_2)}}{e^{ip_3(x - y)} e^{ip_4(x - y)}}$$

$$d^4y d^4x \rightarrow \delta^4(-p_1 + p_3 + p_4) \int d^4(p_2 - p_3 - p_4)$$

$$p_3 = p_1 - p_4$$

$$\Rightarrow \delta^4(p_2 - p_1) \Rightarrow p_1 = p_2$$

$$p_4 \equiv k$$

$$\tau_1 = \frac{-g^2}{2} \int \frac{d^4p_1 d^4p_2 d^4k}{(2\pi)^{12}} e^{ip_1 x_1} e^{ip_2 x_2} \times \left(\frac{i}{p_1^2 + i\epsilon} \right) \left(\frac{i}{p_2^2 + i\epsilon} \right) \left(\frac{i}{(p_1 - k)^2 + i\epsilon} \right) \left(\frac{i}{k^2 + i\epsilon} \right) \delta^4(p_1 - p_2)$$

USE LSZ:

$$\langle f | S | i \rangle = \frac{(-i) \int d^4 x_1 e^{-i p_1 x_1} \partial_1^2}{\langle \Omega | T \phi_1 \phi_2 | \Omega \rangle} (-i) \int d^4 x_2 e^{i p_2 x_2} \partial_2^2$$

$$= - \left(\int d^4 x_1 e^{-i p_1 x_1} (\partial_1)^2 \int d^4 x_2 e^{i p_2 x_2} (\partial_2)^2 \right) \mathcal{L}_1 + \dots$$

$$\int d^4 x_1 e^{-i x_1 (p_i - p_f)} \int d^4 x_2 e^{-i x_2 (p_2 - p_f)}$$

$$= (2\pi)^4 \delta^4(p_i - p_f) (2\pi)^4 \delta^4(p_2 - p_f)$$

$$\mathcal{L}_1 = -\frac{\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{i}{(p_i^2 + i\epsilon)} \right) \left(\frac{i}{p_f^2 + i\epsilon} \right) \left(\frac{i}{(p_i - k)^2 + i\epsilon} \right) \frac{1}{k^2 + i\epsilon}$$

$$\delta^4(p_i - p_2) \times \underbrace{(p_i^2 + i\epsilon) (p_f^2 + i\epsilon)}$$

$\sim$ terms from LSZ.

$$\therefore \langle f | S | i \rangle \sim -\frac{\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{i}{(p_i - k)^2 + i\epsilon} \right) \left(\frac{i}{k^2 + i\epsilon} \right) (2\pi)^4 \delta^4(p_i - p_f)$$

$$\text{Recall, } S = \mathbb{1} + (2\pi)^4 \delta^4(\sum p_i) i\mathcal{M}$$

$$\therefore i\mathcal{M} = \frac{-\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{i}{(p_i - k)^2 + i\epsilon} \right) \left(\frac{i}{k^2 + i\epsilon} \right) + \dots$$

* (i) Internal lines get $\frac{i}{p^2 - k^2 + i\epsilon}$

(ii) Vertices: $i\lambda$

(iii) lines connected to external pts do not get props: cancelled by LSZ.

(iv) Integrate over undetermined momenta

$$\text{ex: } \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \psi \not{D} \psi - \frac{g}{2} \phi^3$$

• $\phi\phi \rightarrow \phi\phi$ Scattering
in the C.O.M frame

$$\frac{d\sigma}{d\Omega}(\phi\phi \rightarrow \phi\phi) = \frac{1}{64\pi^2 E_{cm}^2} |M|^2$$

incoming: p_1^u, p_2^u
outgoing: p_3^u, p_4^u

$$iM_s = \begin{array}{c} p_1 \\ \diagdown \\ p_2 \end{array} \begin{array}{c} \diagup \\ p_3 \\ \diagdown \\ p_4 \end{array} \begin{array}{c} p_1 + p_2 \\ \diagup \\ p_3 + p_4 \end{array} = (ig) \left(\frac{1}{(p_1 + p_2)^2 - m^2 + i\epsilon} \right) (ig)$$

s

$$= \frac{-g^2}{(p_1 + p_2)^2 - m^2 + i\epsilon}$$

$$iM_t = \begin{array}{c} p_1 \\ \diagdown \\ p_2 \end{array} \begin{array}{c} \diagup \\ p_3 \\ \diagdown \\ p_4 \end{array} \begin{array}{c} p_1 - p_3 \\ \diagup \\ p_2 - p_4 \end{array} = (ig) \frac{1}{(p_1 - p_3)^2 - m^2 + i\epsilon} (ig)$$

$$= \frac{-g^2}{(p_1 - p_3)^2 - m^2 + i\epsilon}$$

t

$$iM_u = \begin{array}{c} p_1 \\ \diagdown \\ p_2 \end{array} \begin{array}{c} \diagup \\ p_3 \\ \diagdown \\ p_4 \end{array} \begin{array}{c} p_1 - p_4 \\ \diagup \\ p_2 - p_3 \end{array} = \frac{-g^2}{(p_1 - p_4)^2 - m^2 + i\epsilon}$$

u

$$\frac{d\sigma}{d\Omega}(\phi\phi \rightarrow \phi\phi) = \frac{g^4}{64\pi^2 E_{cm}^2} \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right)^2$$

Mandelstam Variables:

$$s \equiv (p_1 + p_2)^2 = (p_3 + p_4)^2$$

$$t \equiv (p_1 - p_3)^2 = (p_2 - p_4)^2$$

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2$$

which satisfy:

$$\begin{aligned} S + t + u &= p_1^2 + 2p_1p_2 + p_2^2 \\ &+ p_1^2 - 2p_1p_3 + p_3^2 \\ &+ p_1^2 - 2p_1p_4 + p_4^2 \end{aligned}$$